# BIO-301 MCQs MIDTERM- 100%

# GUESS PAPER 2023

## Essentials of Genetics

## BY SULMAN ALI

## 0341-7547044

# OBJECTIVES

1. An allele with only sometimes produce the **symptoms or trait** with which it has been associated at a **detectable level** = **Incomplete dominance**
2. A smooth curve is property of = **Continuous variation**
3. According to Mendel **every trait** is controlled by = **Two factors/alleles**
4. **Skin color of human** is controlled by = **Multiple genes**
5. **Gene density** is more in = **Eukaryotic genomes**
6. **Groups of human chromosomes** are = **23 pairs**
7. **Linkage mapping of Drosophila** was possible **if eye color gene** was placed = **On the X chromosome**
8. **Fur color in Labrador retriever** is controlled by = **Multiple gene**
9. Which of the following is not a **cause of epigenetics**? = **Change in DNA sequence**
10. **Visible regions on chromosomes** are regions of **crossing over = Chiasmata**
11. **Building blocks of chromosomes** are known as **nucleotide genetic helps us to** understand the process of = **Chromosome structure and function**
12. Have the **same length,** shape, centromere location, and gene = **Homologous chromosomes**
13. In **Drosophila**, the **female gene** are located on the = **X chromosome**
14. **Lamp brush chromosomes are found inside** = **Oocytes and spermatocytes**
15. Entire **DNA** of each of the **chromosomes** replicates. Replication is = **Semi-conservative**
16. **First phase of mitosis** = Prophase
17. Example of a diploid cell = **Somatic cell**
18. Meiosis in male is called = **Spermatogenesis**
19. The **two replicated** homologous chromosomes separate during the = **Anaphase I**
20. **Type of cell division** important for **tissue regeneration** is = **Mitosis**
21. Whose **experimental** result gave the **first solid evidence** indicating that a gene is associated with specific chromosomes = **Thomas Hunt Morgan** MCQs:
22. Size of chromatin fiber = **30 nanometers**
23. Dark constriction in chromosome called = **Centromere**
24. **Size of bands in giant chromosome** in Drosophila = **5000** base pairs
25. Building blocks of chromosomes = **Nucleotides**
26. Mendel describe the idea about = **Heredity**

27. **Studies group heredity for traits** determined by **many genes** simultaneously =
28. **Quantitative genetics**
29. The microtubules that push the spindle poles apart are called **aster microtubules**
30. Female bee sex is = **diploid**)
31. Which is correct = 70% are mono-allelic, they exist only in one form.
32. **Height** of human being is an example of = **Continuous variation** MCQs:
33. Condition in which both **alleles** are expressed = **Co-dominance**
34. Color of hydrangea plant changes by soil = **ph level Environmental factor**
35. **XO system of sex present** in = **Insects**, such as **grasshoppers and crickets**
36. DNA replication is = **Semi-conservative**
37. **Single duplication** of chromosomes occurs in = **S-phase of interphase**
38. **If the gene is closely linked, then** = They are less likely to be separated by **recombination events during meiosis**
39. Linkage mapping of **Drosophila** was possible if **eye color gene** is placed = **On the X chromosome**
40. **Gene will be switched on when** = Transcription factors bind to the **promoter region of the gene**
41. Position of the **dark-staining areas** = **Centromere and heterochromatin**
42. Process that converts **the cell into two** is called = **Cell division**
43. **Inherited variation** results from the activity of = **Genes and the environment**
44. **Condensation of chromosomes** is observed in = **Prophase of mitosis/meiosis**
45. Example of **diploid** cell = **Somatic** cell
46. **Genetics helps us in =** Understanding inheritance patterns and **genetic disorders,** developing new technologies and therapies, and studying evolution and diversity
47. **Correct sequence of bottom mapping techniques** = Comparative, physical, linkage, restriction fragment length polymorphism (**RFLP**), amplified fragment length polymorphism (**AFLP**)
48. In **complete dominance, one allele completely** masks the expression of the **other allele.**
49. **Phenotype** depends upon a **complex interaction** between genotype and environment.
50. **Type of heterochromatin** that can change to euchromatin is **Facultative heterochromatin.**
51. **Cytokinesis** in plants occur by **cell plate.**
52. **Sister chromatids** are identical in **mitosis and meiosis II.**
53. **Height of human** beings is an example of **continuous variation**.
54. **Hemophilia** is a **recessive** gene disorder, **X-linked.**
55. In **metaphase**, there is no **RNA** or **protein** synthesis, only **DNA is condensed** and **lined up at the equator of the cell.**
56. **Karyotyping** is the sub-discipline of genetics that deals with the study of the banding pattern of chromosomes.
57. When **Mendel** crossed a true-breeding pea strain that produced smooth seeds in the parental generation, the results were completely dominant to wrinkled seeds.
58. The law of independent assortment with dihybrid gives a genotypic ratio in F2 as **9:3:3:1.**
59. **Variations** that result from acquired characteristics are not acted upon by natural selection and thus do not affect evolution.
60. There are approximately 5,800 **genes present in yeast.**
61. The **allele** is said to have complete penetrance if all individuals who have the disease-causing mutation have clinical symptoms of the disease.
62. During **mitosis**, the centromere that is shared by sister chromatids must divide so that chromatids can migrate to opposite poles.
63. The point mutation in the **CDE III** subdomain completely eliminates the ability of the centromere to function during segregation.
64. **Nucleotides** are the building blocks of nucleic acids.

65. The process by which **prokaryotes** divide their cells into two daughter cells is called binary fission. 19) The microtubules that push the spindle poles apart are called aster **microtubules**.
66. **Meiosis** involves **two cell division** processes, meiosis I and meiosis II, resulting in the formation of four haploid daughter cells from a diploid parent cell.
67. The cytokinesis in animal cells is characterized by a **cleavage furrow.**
68. **CGG** repeats of **200** is seen in individuals with fragile X.
69. The principles of heredity were explained by **Gregor Mendel**.
70. The cross of TTPP x ttpp would result in the F1 generation being **TtPp**.
71. A correct statement about Mendel's model is that there is a clear dominant-recessive **relationship between alleles**.
72. When the centromere is very near to the end of the chromosome, it is called **telocentric**.
73. There are four types of chromosomal **bands in humans**.
74. In prokaryotes, a **single circular DNA i**s present in the nucleoid region.
75. **Chromosomes** are identified by their size, shape, and banding pattern.
76. A ZW female is **heterogametic**.
77. **Lampbrush chromosomes** are found in all **vertebrates, invertebrates, and plants**.
78. **Lampbrush** chromosomes were discovered by **Flemming in 1882.**
79. As a **result of mitosis,** the cells produced are called **daughter cells.**
80. The **X-O system** is present in **grasshoppers**.
81. If a **normal father** and **color-blind mother** have children, **the ratio of blind sons** will be **2:4**.
82. The **duration of interphase** is **90%** of the **cell cycle**.
83. The **nutrition of a mother** can affect the **epigenome** of a **fetus**.
84. **Two enzyme types** are involved in **histone acetylation**.
85. When **two mutations** together have a smaller effect than the effect seen when they are alone, it is called a **negative effect.**
86. In the **ABO blood group system**, two of the alleles are **codominant** to each other.
87. If a **mother** has a **disorder**, she will **transmit** the **mutant allele** to her **sons** but not **her daughters.**
88. In the ZW and ZZ system, the **female is ZW.**
89. The **70nm** is the **relative frequency** of a **particular** characteristic.
90. In **telophase of meiosis**, the cells divide into **four haploid cells**.
91. The G0 phase is also called the **post-mitotic phase.**
92. **Reproduction in prokaryotes** occurs through **binary fission.**
93. The **building blocks of proteins** are **amino acids.**
94. In E.coli, approximately **3000-4000 genes** are present.
95. A **point mutation** in the **CDE III** subdomain **completely eliminates** the ability of **centromere** function.
96. **Linker DNA** has approximately **20-60 base pairs.**
97. The **comb shape** in chickens can exhibit **four different shapes.**
98. E.coli has approximately **3000-4000 genes** organized into its one circular chromosome.
99. The **comb shape** is controlled by **two genes** found on **two different chromosomes.**
100. The genotype **bbee** is not of the **golden cast color in Labrador retrievers.**
101. **Co-dominance** appears in the **AB blood group phenotype.**
102. **A female bee's sex** is determined **by the ZW system.**
103. **Tetrads align** at the center in **metaphase**.
104. The **pleiotropic effect** is shown in all of **cystic fibrosis, sickle cell anemia**, and **Huntington's disease.**
105. The ratio of F2 in a **dihybrid** cross between TTRR and ttrr is **9:3:3:1.**
106. **Protozoa** can have **300** or more **chromosomes**.
107. **Roundworms** have only **two chromosomes**.
108. **Onions** and **molds** have **16** chromosomes.
109. **Skin color pigmentation in humans** is controlled by **at least three** separately inherited genes.

110. **Proteins** that are used to **package DNA** are called **histones**.
111. **Offspring of the same variety** are termed as a **true breed**.
112. **Variations** that are genetically **controlled** cannot be altered.
113. **Gregor Johann Mendel** is known as the **father of genetics.**
114. Which do not follow principle of heredity formulated by **Gregor Mandel**__

**Epistatis**

115. A **gene's capacity to impact an organism** in a variety of ways is termed as ___
116. **Pleiotropy**
117. When alleles at **two gene loci** segregate **together and transmitted as a unit**, what
118. Form of segregation occurs__ **Dependent segregation**
119. ___ is **heredity unit** of chromosomes__ **Gene**.
120. A **DNA** polymorphism affects ___ of the population__**1-2%**
121. population in human is controlled by… **3** …**genes**.
122. Number of genes in drosophila are…. **13700**
123. Chromosome with **same** length shape size …. **Autosome**
124. Which band correspond to **constitutive heterochromatin**…**C band**
125. DNA base pair length is … **120** BP
126. Prokaryotes simply Divide their cells into Two daughter cells by ……**. Binary Fission**.
127. In animal cell cytokinesis divide the cell…. **Cytoplasm**
128. 80)**ph of soil** will change the color of____ hydrangea flower.
129. Chromosomes found in trillium are .. **32micro**
130. ___Takes the 90% time of cell division. **Interphase**
131. **presence of Y chromosome** determines **maleness** in most organisms, including **humans**.
132. Cross of Gg green color and Yy yellow color produces **….(F1 generation of**
133. **GgYy** )
134. Which statement is incorrect according to prokaryotes**….( Prokaryotes have**
135. **Low genetic diversity than eukaryotes**)
136. The process that divide cells into two cells **… Cell division**
137. **Diameter** of DNA___ **2nm**
138. ___Studie**s group Hereditary for traits Determined by many Genes**
139. Simultaneously__ :**Quantitative genetics**
140. The change in the flower color of hydrangea plant is due to the **pH of soil**.
141. less than **100%** penetrance is a result of **modifiers**, **epistasis**, **suppressor**, and **environmental conditions**.
142. **Five types of histone molecules** are found in nature.
143. Chromosomes are identified by **number, size, and banding pattern.**
144. The other name of **polytene chromosome** is **salivary gland chromosome.**
145. **23 pairs** of homologous chromosomes are found in humans.
    - blood **group** is a **universal donor**.
146. Until the 19th century, biologists believed that inheritance was **blending**.
147. **One map unit (1 cM)** is equivalent to a **1% recombination frequency.**
148. **Chargaff's rule** states that **A** always pairs with **T**, and **G** always pairs with **C** in DNA.
149. In **metaphase**, chromosomes are attached to kinetochores on the **spindle fibers**.
150. **M phase** (mitosis or meiosis) is not present in **interphase**.
151. The number of chromosomes is **halved** in **meiosis**.
152. The statement "yy produce yellow color" is incorrect. **Homozygous recessive individuals (yy) produce green color in plants.**
153. The **longest chromosome size** is about 30 micrometers (**30 µm)**.
154. The **behavior of chromosomes** during cell division is according to Mendel's laws of **segregation** and **independent assortment.**
155. The **number of chromosomes** in eukaryotes is **multiple** and varies among different species.

156. **Hyper methylation** occurs at **CpG dinucleotides** and results in silencing of genes, such as the **fragile X chromosomes.**
157. **Mendel's laws** include the law of dominance**, law of independent assortment, and law of segregation.**
158. The **transfer of blood** is known as **blood transfusion.**
159. The **causes of epigenetics** include **histone modification**, **DNA methylation, and miRNA regulation.**
160. Male bee sex is **haploid (n**).
161. The **gene of green pepper** can be represented by the alleles rr (for red color) and yy (for yellow color).
162. **Quantitative genetics** studies heredity for traits determined by **many genes** simultaneously.
163. The **population in humans** is controlled by **3** genes.
164. Drosophila has around **13700 genes.**
165. **Autosomes** are chromosomes with the same length and shape size.
166. **C band** corresponds to constitutive **heterochromatin.**
167. DNA base pair length is **120 bp.**
168. **Prokaryotes** simply divide their cells into two daughter cells by **binary fission.**
169. In animal cells, cytokinesis divides the cell through a **cleavage furrow.**
170. The **nutrition of a mother** can affect the epigenome of the **fetus**.
171. **CGG** repeats are seen in individuals with **fragile X syndrome**.
172. **Histone acetylation** is a process that involves the addition of **an acetyl group** to a lysine residue on a **histone protein**.
173. **Epigenetics** is the study of changes in gene expression that occur without **changes** in the **DNA sequence.**
174. When **two mutations** together have a smaller effect than their effect when alone, it is called a **negative epistasis**.
175. In the **ABO blood group system**, the two alleles are **co-dominant** to each other.
176. A **mother with a disorder** will transmit the mutant allele to all sons but not daughters if the **disorder is X-linked.**
177. In **ZW and ZZ systems,** the female genes are **ZW**.
178. **70 nm** is the character of the **relative frequency in a population.**
179. **In telophase,** the cells are divided into **four haploid cells in meiosis.**
180. **G0 phase is also called the post-mitotic phase.**
181. **Reproduction** in prokaryotes is through **binary fission.**
182. Amino acids are the building blocks of **proteins**.
183. In E.coli, about **3000-4000 genes** are present.
184. A point mutation in the **CENP-A** sub-domains completely eliminates the ability of the centromere to function.
185. Linker DNA has **20-60 base pairs.**

186. **Co-dominance** shows **incomplete dominance in inheritance.**
187. **9:3:3:1** is the phenotypic ratio of the F2 generation in a **dihybrid** cross between TTRR and ttrr.
188. The **number of chromosomes in protozoa** varies depending on the species.
189. **Geneticists and researchers** can study genetics in a reverse pattern in reverse genetics than in forward genetics, where the mutant gene is identified based on its phenotype.
190. **Skin color pigmentation** in humans is controlled by at least **three** separately inherited genes.
191. The **change in flower** color of the hydrangea plant is due to changes in **soil pH**.
192. **Less than 100%** penetrance is a result of environmental factors affecting gene expression.
193. **Five types of histone molecules** (H1, H2A, H2B, H3, and H4) are found in nature.
194. **Chromosomes** are identified by their **size, shape, and banding pattern.**

**195.** **Polytene chromosomes** are also known as giant chromosomes.

**196.** Human cells contain **23 pairs** of **homologous chromosomes**, making a total of **46 chromosomes.**

**197.** The blood group O is a **universal donor.**

**198.** **Epistasis** is the interaction between two or more genes where one gene masks or modifies the **expression of another gene.**

**199.** Until the **19th century,** biologists believed that inheritance was a **blending of traits from both parents.**

**200.** (**200) CGG repeats** are seen in individuals with **fragile X.**

**201.** The principles of heredity were explained by **Gregor Mendel.**

**202.** In a cross between **pure-breeding TTPP** and ttpp, the F1 generation obtained is **TtPp**.

**203.** A clear dominant-recessive relationship between alleles is the correct statement about **Mendel's model.**

**204.** **Centromere** is very near to the end in **telocentric chromosomes.**

**205.** **Four types of chromosome band**s are present in humans.

**206.** **Centromere** is the constricted region of a chromosome that holds **sister chromatids together.**

**207.** Chromatids are the **two identical copies** of a chromosome that are joined at the centromere.

**208.** In prokaryotes, **single circular DNA** is present in the **nucleoid**.

**209.** **Chromosomes** are identified by their size, shape, and banding patterns.

**210.** **ZW female heterogametic** refers to the sex determination system in birds and some other species where females have one Z and one W chromosome.

**211.** **Lampbrush chromosomes** are found in vertebrates, invertebrates, and plants.

**212.** The cells produced as a result of mitosis are called **daughter cells**.

**213.** **X-O system** is present in grasshoppers for sex determination where females have two X chromosomes and males have one X and one O chromosome.

**214.** …. Normal Father Color Blind Mother.. sons will be ratio.

**215.** Option ya thi..

**216.** **¼**

**217.** 2/4

**218.** ¾

**219.** 4/4

**220.** which is the Population genetics is the study In a group of population.?

- Gene b) heredity c) trait d) **genome**

**221.** **Quantitative genetics** is the study of group hereditary for traits determined by many genes simultaneously such as.. Skin color, Height, eye color **d) all of these**

**222.** Molecular genetics this branch of genetics deals with the molecular structure and function of genes **Gene**

**223.** What is Trait?

**224.** Ans; variant for a character (i.e. brown, black, white etc).

**225.** Define Genotype and Phenotype? Genetic make-up of an organism is called Genotype while physical appearance of an Organism is called Phenotype

**226.** When one characteristic expresses itself over the other i.e. round over wrinkled was dominant in Gregor Mendel experiments while the trait that does not show through in the first generation is called as **recessive trait** i.e. wrinkled. when one characteristic expresses itself over the other i.e. round over wrinkled was Dominant in Gregor Mendel experiments-IS-Called **trait**

**227.** when the trait that does not show through in the first generation is called as i.e. wrinkled.? **Recessive trait**

**228.** when we a cross of parents, that are pure for contrasting traits, only one form of the Trait will appear in the next generation. In the monohybrid cross, one version will be disappeared is called **law of dominance**

**229.** In which process do we describe the situation in which there are many intermediates between the extremes, such as shades of hair color between black and blond? **Continuous variation**

**230.** Which process is entirely genetically controlled, and traits cannot be altered by external conditions? Blood group, color blindness, and dwarfism are examples of these types of traits. **Discontinuous Variation**

**231.** In which process do tall plants produce only tall offspring, and purple flowered plants produce only purple offspring? **Pure-breeding plants**

**232.** What is the pattern of inheritance that deviates from the Mendelian pattern of inheritance? Non-Mendelian inheritance

**233.** Each trait is controlled by how many genes? **Single gene**

**234.** How many alleles does each gene have? **2**

**235.** In a cross of true-breeding strains, how many of the parents do the offspring resemble? **One**

**236.** Which law states that two members of a gene pair segregate (separate) from each other during the formation of gametes? **Law of segregation**

**237.** True or False: In incomplete dominance, the phenotype of the heterozygous offspring is a blend of the phenotypes of the two homozygous parents. **True**

**238.** What term refers to the genetic makeup of an individual? **Genotype**

**239.** What is a change in the genetic material of an organism called? **Mutation**

**240.** Which of the following is NOT a type of gene mutation? **Translation**

**241.** Translation (Translation is a process in protein synthesis and not a type of gene mutation)